



Anterior cruciate ligament reconstruction combined with anterolateral ligament reconstruction using hamstring autograft versus anterior cruciate ligament reconstruction using bone-patellar tendon-bone autograft: a randomised controlled trial with 5-year follow-up

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Summary

Background Anterior Cruciate Ligament (ACL) rupture is common knee injury. Although ACL reconstruction (ACLR) is standard, graft failure rates remain high in young active patients. This study investigated whether combining ACLR with anterolateral ligament (ALL) reconstruction (ALLR) reduces grafts failure compared with ACLR.

Methods In this prospective, single-centre, randomised controlled trial conducted at the Santy Orthopedic Center in Lyon, France, patients aged 18–35 years with symptomatic ACL rupture were randomly allocated (1:1) to ACL + ALL reconstruction using hamstring tendon autograft (ACLR + ALLR) or ACLR with bone-patellar tendon-bone autograft (ACLR). Randomisation was performed with a block size of four using telematic software by an independent study coordinator, with concealed allocation. Surgeons were informed of the assigned procedure on the morning of surgery. Outcome assessors were not blinded. The primary outcome was graft failure at 5 years, assessed clinically and by magnetic resonance imaging (MRI) by an independent sports medicine physicians not involved in the index surgery. Efficacy analyses were performed on the Full Analysis Set in accordance with the intention-to-treat principle, while safety analyses were conducted on the Safety Set. Trial registration: [ClinicalTrials.gov](https://clinicaltrials.gov), ID NCT03740022. The trial has been completed.

Findings Between November 11, 2016, and January 20, 2020, 593 patients were randomized (297 assigned to ACLR + ALLR and 296 to ACLR). The mean age was 25.0 years (SD 4.5); 447 (75%) participants were male and 146 (25%) female. Of these 593 patients, 556 (94%) completed a mean 5-year follow-up. Graft failure occurred in 12/283 (4.2%) with ACLR + ALLR versus 28/273 (10.3%) with ACLR ($p = 0.006$; adjusted odds ratio 2.54 [95% CI 1.27; 5.36]— $p = 0.008$). The number needed to treat was 17 overall, and 9 in patients younger than 25 years.

Interpretation In our study of young, active adults with ACL rupture, who are considered high-risk for graft failure, combining ACL reconstruction with anterolateral ligament reconstruction (ACLR + ALLR) significantly decreased graft failure compared with ACLR. These results suggest that ACLR + ALLR might be beneficial for young or highly active individuals and provide a basis for future research to refine patient selection, evaluate long-term outcomes beyond five years, and explore benefits in other subgroups of patients with ACL injuries.

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Keywords: ACL; Graft failure; LEAP; Reoperation rate; Clinical Outcomes; Randomised controlled trial

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Research in context

Evidence before this study

We searched PubMed from inception to 29 October 2025 using terms including “ACL rupture,” “ACL reconstruction,” “graft failure,” “anterolateral ligament,” and “lateral extra-articular tenodesis,” with no language restrictions. Clinical trials, cohort studies, and biomechanical studies reporting graft failure or knee stability outcomes were analysed. Anterior cruciate ligament (ACL) rupture is a common sports injury, and ACL reconstruction (ACLR) is the standard of care for patients with symptomatic instability. However, graft failure after ACLR is a concern, particularly in young adults (18–35 years), with higher rates in active populations. Despite decades of technical advancements, failure rates have not substantially declined in high-risk populations. Biomechanical and clinical studies have suggested that lateral extra-articular procedures (LEAPs), including anterolateral ligament reconstruction (ALLR), may enhance joint stability and reduce the risk of graft failure. High-quality level I evidence for LEAPs was limited prior to this study. The STABILITY trial showed improved outcomes with modified Lemaire tenodesis (a non-anatomical LEAP), but did not compare against bone-patellar tendon-bone (BPTB) autograft or evaluate anatomic ALLR.

Added value of this study

To the best of our knowledge, this randomised controlled trial (RCT) is the first to compare BPTB graft ACLR—commonly used in young, high-risk adults—against ACLR combined with a lateral extra-articular procedure (LEAP). It is also the first to specifically evaluate ALLR which is a LEAP already in common clinical practice. Furthermore,

the 5 year follow provides the first mid-term RCT data using a contemporary surgical and rehabilitation technique. The major new findings are that ACLR + ALLR significantly reduces graft failure rates at five-year follow-up. The addition of ALLR resulted in a statistically significant reduction in graft failure compared with BPTB ACLR. Overall reoperation rates were also significantly lower in the combined group, largely due to differences in graft rupture rates but also due to significantly higher rates of cyclops syndrome and secondary meniscectomy in the BPTB group. Functional outcomes, including patient-reported measures and return to sport, were similar between groups, indicating that the combined procedure does not impair recovery. The study addresses an important knowledge gap by evaluating ALLR in the context of high-risk populations, providing a rigorous assessment of its potential protective role in ACL reconstruction.

Implications of all the available evidence

These findings challenge the widely held belief that ACLR performed with a BPTB graft is what some refer to as the gold-standard choice for surgical treatment of young, active adults at elevated risk of graft failure. This is because combined ACLR and ALLR significantly outperforms BPTB graft ACLR with respect to graft rupture rates and overall reoperation rates. The addition of an anterolateral procedure confers protective biomechanical benefits by better restoring native knee kinematics and reducing ACL graft forces. Taken together with prior biomechanical and retrospective clinical studies, this trial provides strong evidence for a shift in ACLR strategy towards combined reconstructions in high-risk populations.

Introduction

Anterior cruciate ligament (ACL) ruptures are amongst the most common sports injuries, and it is estimated that approximately 150,000 anterior cruciate ligament reconstructions (ACLR) are performed annually in the United States.^{1,2} Although overall failure rates are low, graft failure remains a major concern in high-risk populations. Re-injury rates can exceed 20%, especially in younger patients who return to jumping, pivoting, or cutting sports.^{3,4} ACL graft failure has significant clinical implications, including an increased risk of subsequent meniscal injury and chondral damage, potentially leading to accelerated joint degeneration, poor functional outcomes and increased healthcare costs.⁵ Although non-operative treatment or primary ligament repair may be considered in selected situations, ACL reconstruction remains the standard of care for young active adults.^{6,7}

Despite decades of advancements in surgical techniques, graft rupture rates after ACLR remain a persistent concern,⁸ particularly in high-risk

populations. Recent comparative studies,^{9–13} including the RCT STABILITY trial,¹⁴ demonstrate that adding a lateral extra-articular procedure (LEAP) to ACLR significantly reduces graft failure rates (4% versus 11% at 2 years, using hamstring autografts, compared to ACLR) and reoperation rates by improving knee stability.^{15,16} Recent meta-analyses confirm these findings, reporting a pooled 50–60% reduction in graft reinjury rates with LEAPs without increased complications.^{17,18} The biomechanical rationale is that LEAPs share load with the ACL graft, better restoring knee kinematics.¹⁹ However, limited level I evidence has hindered wider LEAP adoption, necessitating this study to compare ACLR with BPTB autograft to ACLR + ALLR in a young active adults.

The aim of this study was to compare the graft failure rates following ACLR performed with the commonly used graft choice of bone-patellar tendon-bone (BPTB) autograft against combined ACLR and ALLR using hamstring tendon autografts in a young active adults.

Methods

Study design

The trial was an investigator-driven, single-center, prospective, randomised, controlled clinical trial performed at Santy Orthopedic Center. This trial was performed in accordance with the declaration of Helsinki and European guidelines for Good Clinical Practice. All the patients provided written informed consent before inclusion in the study. Approval was granted by the relevant ethics committee, Comité de Protection des Personnes Sud-Est III, on 1st March 2016 (registration number: 2016-007B). The study was registered on [ClinicalTrials.gov](https://clinicaltrials.gov) NCT03740022. The original protocol specified a 3-year primary endpoint for graft failure; however, a formal protocol amendment (MS number: 3, dossier number: 16.00552.000007, approved 12 April 2022) extended the follow-up to 5 years due to disruptions from the COVID-19 pandemic affecting surgical practices, patient inclusion, and return-to-sport outcomes.

Participants

Patients eligible for enrolment in the trial were men and women, between 18 and 35 years of age, who presented with an ACL rupture, had symptomatic knee instability, and took part in sports activities at least once per week.

Patients who presented with multi-ligament injuries (posterior cruciate, medial collateral and lateral collateral ligaments), had a BMI under 18.5 or over 30 kg/m², took part in professional sport, had a recurrent ACL rupture, had a history of contralateral knee ligament injury, had connective tissue disorders, congenital conditions, and chondral lesions requiring treatment other than chondroplasty were excluded from the study.

Sex was recorded as biological sex (male or female) at enrollment, based on participants' medical records.

Randomisation and masking

Participants were randomly assigned to one of the two surgical techniques in a 1:1 ratio, with a block size of 4, by a clinical research assistant using telematic software (CSRandomization module of ENNOV Clinical) to generate the allocation sequence. The allocation list was concealed and securely stored at the centre. On the morning of surgery, each surgeon was informed of their allocated procedure.

Outcome assessors could not be blinded because of the different scar types. The primary outcome was assessed clinically and by magnetic resonance imaging (MRI) by independent sports medicine physicians who were not involved in the index surgery, in order to minimise bias.

Trial oversight and safety monitoring

No formal independent safety monitoring board was established for this single-centre, investigator-initiated

trial. However, data and safety oversight were ensured through multiple independent entities not involved in patient care: Ascopharm Groupe Novasco (acting as the regulatory sponsor) monitored study data, C2R Épidémiologie served as an external data management centre, and Horiana Health Data Consulting performed the independent statistical analysis. Patient safety was monitored by a clinical research assistant at each scheduled follow-up (3 weeks, 6 weeks, 3 months, 6 months, 1 year, 3 years, and 5 years postoperatively) or as needed if the patient required additional evaluation by the sports surgeon.

Procedures

Surgeries were performed by one of the three senior orthopaedic surgeons in a standardized manner in accordance with previous technical descriptions.^{20,21} In summary, in the control group (ACLR), a BPTB autograft was used. A 10-mm BPTB graft was harvested with a 9 × 15-mm patellar bone plug and an 11 × 20-mm tibial bone plug. The femoral tunnel was drilled in an outside-in fashion. The graft was passed through the femoral tunnel in an antegrade fashion. The graft was fixed in the femoral tunnel using a press-fit technique and then fixed in the tibial tunnel with the knee flexed at 30° using an interference screw (BioComposite interference screw: Arthrex).²⁰ For the ACLR + ALLR group, hamstring tendon autografts were used to create both the ACL and ALL grafts. The semitendinosus and gracilis tendons were harvested using an open-ended tendon stripper. The tibial insertion was preserved to improve fixation and vascularity. A combined 4-strand ACL graft and single-strand ALL graft was prepared using a tripled semitendinosus tendon with an additional length of gracilis tendon sutured to it. A single femoral tunnel was drilled in an outside-in fashion. Intraarticularly, this was placed in a mid-anteromedial bundle position; laterally, it was positioned at the femoral origin of the ALL (proximal and posterior to the lateral epicondyle). The ACL portion of the graft (3-part semitendinosus tendon and 1-part gracilis tendon) was fixed on both sides using bioabsorbable screws (BioComposite interference screw: Arthrex) with the knee flexed to 30°. The additional length of gracilis tendon that emerged from the femoral tunnel at the lateral cortex formed the ALL portion of the graft. This was passed under the iliotibial band using a suture grasper, through a tunnel in the proximal tibia, and then back to the ALL origin, where it was tensioned and fixed with the knee in extension, completing anatomic ALLR (Fig. 1A and B).²¹ In both procedures, the intra-articular tunnel placement was standardized to ensure the same anatomical position. For meniscal repairs, a posteromedial accessory portal with a 25° hook (SutureLasso; Arthrex) loaded with a No. 0 absorbable monofilament suture (PDS; Ethicon) was used for the medial meniscus and an all-inside

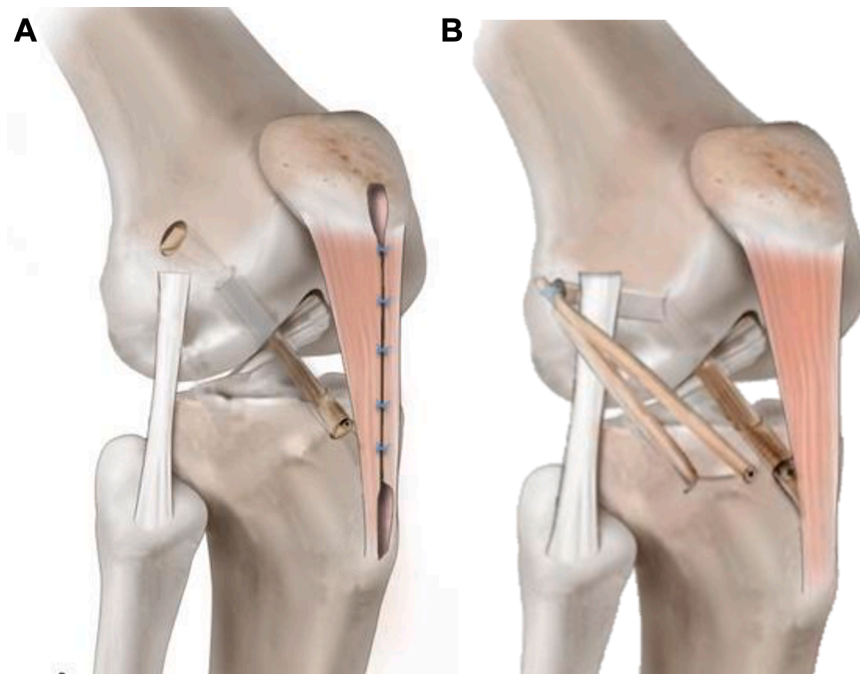


Fig. 1: ACL reconstruction techniques. (A) ACL reconstruction using bone-patellar tendon-bone (BPTB) autograft, reprinted with permission.¹⁰ (B) Combined ACL reconstruction (ACLR) and anterolateral ligament reconstruction (ALLR) using hamstring tendon autografts, reprinted with permission.²²

anchor (Ultra Fast-Fix, Smith & Nephew) was used for the lateral meniscus. All patients followed the same postoperative rehabilitation protocol with immediate brace-free full weight bearing. In cases of meniscal repair, knee flexion was limited to 90° for the first 6 weeks and return to run was permitted at 4 months. Resumption of sports activities was allowed after 3–4 months for nonpivoting sports, 6 months for noncontact pivoting sports, and 8–9 months for contact pivoting sports.

Outcomes

The primary outcome was the occurrence (or not) of graft failure at 5 years $\pm$ 10 months. Graft failure was assessed by history of instability symptoms, physical examination, and MRI. An independent sports medicine physician evaluated knee laxity using the following criteria: Lachman test (positive if soft/no endpoint, >3 mm anterior tibial translation versus contralateral knee) and pivot shift test (positive if graded + [glide], ++ [clunk], or +++ [gross]). Side-to-side anteroposterior laxity was measured using the Rolimeter (Aircast, Summit, New Jersey, USA), performed at 20–30° of knee flexion with manual maximal anterior force, and for each knee, three measurements were obtained, and the mean value was recorded for analysis. Differences >3–5 mm suggested graft failure when combined with clinical findings. All patients with positive physical

examination findings and/or history of instability underwent MRI to confirm the clinical suspicion of graft failure.²²

Graft survivorship was defined as the absence of confirmed graft failure over the study period. Time to graft failure was defined as the time interval between surgery and the last follow-up.

Secondary outcomes were the subjective and objective International Knee Documentation Committee Subjective Knee Form (IKDC),²³ Lysholm Knee Scoring Scale (Lysholm),²⁴ Knee Injury and Osteoarthritis Outcome Score (KOOS),²⁵ Tegner Activity Scale (Tegner),²⁶ return-to-sport rates, side-to-side laxity (Rolimeter, Aircast, Summit, New Jersey, USA), quantitative pivot shift (QPS) measurements, and radiographic evaluation of osteoarthritis progression.

In this manuscript, results are presented for all listed outcomes except postoperative QPS and radiographic evaluations. Postoperative QPS assessments were discontinued due to inconsistencies between preoperative measurements (performed under general anaesthesia) and postoperative measurements (performed in awake patients), which affected measurement reliability. Therefore, only baseline QPS data are reported. Radiographic evaluations were not reported here, as osteoarthritis progression was previously analysed at 10 years in a separate study,²⁷ which found that ACLR + ALLR did not increase the risk of lateral

tibiofemoral osteoarthritis compared with ACLR at medium-term follow-up.

Post-hoc outcomes included adverse events, reoperations, contralateral ACL ruptures and timing of graft failures. These exploratory analyses were performed to provide additional clinical context beyond the pre-specified secondary endpoints.

Contralateral ACL rupture was defined as a rupture of the ACL in the opposite knee during the 5-year follow-up, identified through clinical evaluation, abnormal laxity (positive Lachman or pivot shift tests), and confirmation by MRI.

Adverse events were defined as complications that did not require reoperation, whereas secondary reoperations were followed by complications requiring surgical intervention; all adverse events, including those not considered direct complications of the index surgery, were systematically recorded at each scheduled follow-up or as needed.

Patient assessments were conducted at the time of inclusion, 3 weeks, 6 weeks, 3 months, 6 months, 1 year, 3 years, and 5 years after surgery. PROMs questionnaires were administered at baseline, 1 year, and the last follow-up visit, with routine postoperative follow-up by our sports medicine team. Although examiners were independent of the surgical team, complete blinding was not feasible because surgical scars revealed the procedure performed. Side-to-side anteroposterior laxity was measured at baseline and during the first year after surgery.

Statistical analysis

A 7% graft failure rate was estimated for the control group at 5-years postoperatively. It was estimated that 592 patients (296 patients per group) would be required for the study to have 80% power to detect a 5% reduction in graft failure rate in the ACLR + ALLR group at a two-sided alpha level of 5%. Recruitment was increased by 10% to account for expected loss to follow-up during the study period.

The full analysis set (FAS) included all randomised patients who underwent surgery and had a primary endpoint available at 5-years follow-up. Safety data were assessed in the safety population, defined as all randomised patients who underwent surgery, regardless of whether they were lost to follow-up during the study period. Patients in this set were analysed according to the surgery received. Efficacy analyses were performed on the Full Analysis Set in accordance with the intention-to-treat principle, while safety analyses were conducted on the Safety Set. Full details of statistical analyses are available in the [Appendix](#). In summary, graft failure rates at 5 years were compared using a χ^2 test and presented with its 95% CI. Multivariate logistic regression was performed including the following covariates: sex (Male versus Female), age (<25 years versus ≥ 25 years), BMI (<25 kg/m² versus ≥ 25 kg/m²),

meniscal lesions (Yes versus No), preoperative Tegner score (≥ 7 versus <7) and type of sport leading to the rupture (Pivot/Contact, Pivot, No Pivot/No Contact, No Sport). Graft survivorship was evaluated using the Kaplan–Meier method. The Log-rank test was used to compare graft survival curves. A Cox proportional hazards model was also used to compare time-to-event distributions, adjusting for the same covariates. The index date for time-to-event analyses was defined as the date of initial surgery.

Subgroup analyses were conducted by adding treatment-by-subgroup interaction terms to the logistic regression model for the primary outcome at 5 years and all clinically relevant factors. Subgroup analyses were exploratory and interpreted cautiously without adjustment for multiplicity.

Exploratory analysis was conducted among patients younger than 25 years. A cutoff of 25 years was selected to define young adults, based on prior literature, including the STABILITY trial,¹⁴ which identified individuals under 25 years as a high-risk group for ACL graft failure due to elevated activity levels and biomechanical demands in younger, active populations.

The minimal clinically important difference (MCID) was defined as half the standard deviation of the change from baseline in the score at 5 years, consistent with distribution-based methods in ACLR studies where anchor-based estimates vary.^{28,29} MCIDs were calculated for subjective IKDC and Lysholm scores. Number needed to treat (NNT) was defined by $1/(\text{Control Event Rate} - \text{Experimental Event Rate})$. The log-rank test and all outcome analyses were conducted at the 5-year follow-up timepoint.

All secondary outcomes analyses were considered exploratory and are detailed in the [Appendix](#).

Statistical analyses were performed using SAS for Windows (version 9.4; SAS Institute Inc.), with significance defined as $p < 0.05$.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

From November 11, 2016, to January 20, 2020, 638 patients were randomised; 45 declined participation, and 593 underwent surgery (297 [50%] to the ACLR + ALLR group using hamstring tendon autografts and 296 [50%] to the ACLR group using bone-patellar tendon-bone [BPTB] autografts) in a 1:1 ratio. In this study, 526 (95%) of patients were well-trained and frequently engaged in sports weekly, with 498 (90%) participating in pivot sports (e.g., soccer, handball, basketball, tennis) and 58 (10%) in non-pivot sports (e.g., running, cycling, swimming).

Fig. 2 shows the study flow diagram: 283 (95%) patients in the ACLR + ALLR group, and 273 (92%) patients in the ACLR group reached the study endpoint by completing the 5 years follow-up assessment (Fig. 2). The mean follow-up after surgery was 61.5 months in the ACLR + ALLR group and 63.9 months in the ACLR group, with a minimal follow-up of 52 months among patients without graft failure for each group. Baseline characteristics of the study population are shown in Table 1. These characteristics were similar between both groups.

Of the 593 randomised participants, all underwent surgery. However, 3 participants randomised to ACLR + ALLR received ACLR, and 8 participants randomised to ACLR received ACLR + ALLR due to intraoperative clinical decisions (e.g., anatomical or tissue-related factors), as documented in the case report forms. The 5 years follow-up (± 10 months) was achieved in 556 (94%) of 593 randomised patients, with 14 (5%) lost in the ACLR + ALLR group and 23 (8%) in the ACLR group. Baseline characteristics of factors included in regression models were similar between patients included in the FAS and lost to follow-up patients (Supplementary Appendix). The Safety Set (SAF, $n = 593$) analysed participants based on actual surgery received, while the Full Analysis Set (FAS, $n = 556$) included only those with 5-year follow-up data, analysed by randomised allocation (283 in ACLR + ALLR, 273 in ACLR).

Primary outcome

For the primary outcome, in the Full Analysis Set (FAS, $n = 556$), a total of 40 (7.2%) graft failures were

observed at 5-year follow-up. The ACLR + ALLR group had 12 graft failures (4.2%, $n = 283$), while the ACLR group had 28 graft failures (10.3%, $n = 273$). The unadjusted Chi-square test showed a significant difference ($p = 0.006$). On multivariate logistic regression, ACLR (OR 2.54 95% CI: [1.27; 5.36], $p = 0.0077$) and age < 25 years (OR 3.08 95% CI: [1.51; 6.74], $p = 0.0017$) were significantly associated with the probability of graft rupture at 5 years, adjusted for sex, BMI, meniscus lesion, Tegner score, sports, and type of sport causing the rupture (Fig. 3).

Similar findings were observed in subgroup analysis of patients younger than 25 years old (7/147, 5% versus 22/134, 16%; 95% CI: [6.8; 13.9], $p = 0.001$). The NNT to prevent one graft failure was 17 overall, and 9 in patients younger than 25 years old. At 6 years, a total of 42 graft failures were observed in the FAS. Kaplan–Meier survival analysis demonstrated higher cumulative graft survivorship in the ACLR + ALLR group (Log-rank $p = 0.003$ [Fig. 4]).

A multivariate Cox proportional hazard model, accounting for potentially important covariates, also demonstrated that ACLR was associated with a significantly greater risk of graft failure compared to ACLR + ALLR (HR 2.65; 95% CI: 1.38–5.42, $p = 0.003$ [Fig. 5]). Age < 25 years (HR 2.81; 95% CI: 1.43–5.93; $p = 0.002$) was also an independent predictor of graft failure included. Sex, BMI, meniscal repair, Tegner score, and sports type causing the rupture were not significantly associated with graft failure. Similar results were observed in patients younger than 25 years old (HR 4.24, 95% CI: 1.91–10.72, $p = 0.0002$).

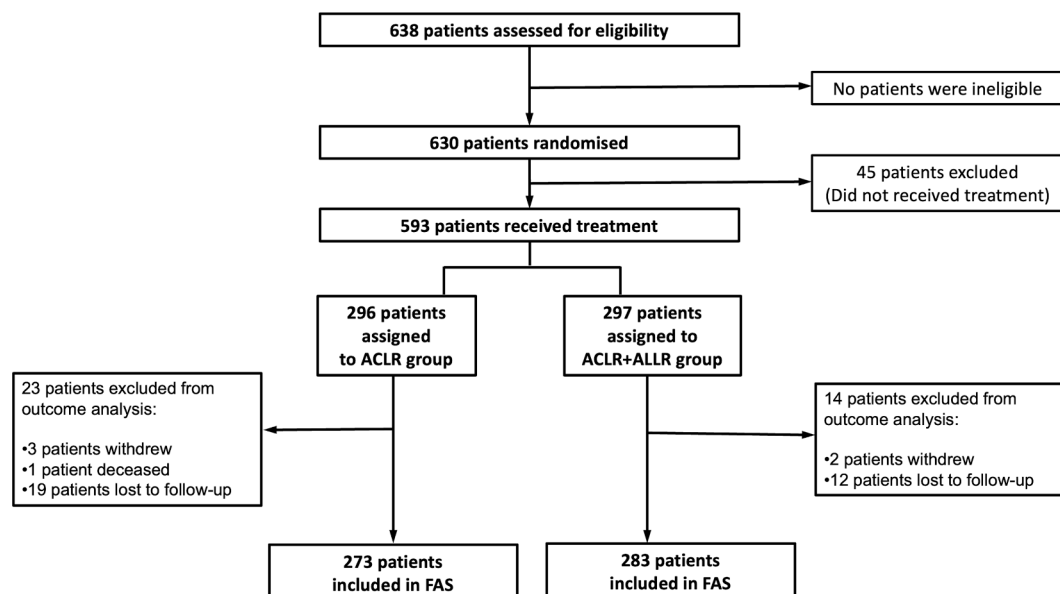


Fig. 2: Trial profile. ACLR, anterior cruciate ligament reconstruction; ALL, anterolateral ligament.

Category	ACLR (n = 273)	ACLR + ALLR (n = 283)
Sex, n (%)		
Male	233 (85.3)	214 (75.6)
Female	40 (14.7)	69 (24.4)
Age (years)		
Mean (SD)	25.2 (4.6)	24.8 (4.4)
Class of age (years), n (%)		
<25	136 (49.8)	145 (51.2)
≥25	137 (50.2)	138 (48.8)
Body Mass Index (kg/m²)		
Mean (SD)	23.8 (2.4)	23.8 (2.5)
Class of BMI (kg/m²), n (%)		
<25	196 (71.8)	205 (72.4)
≥25	77 (28.2)	78 (27.6)
Time from injury to surgery (months)		
Mean (SD)	7.1 (16.2)	6.9 (15.7)
Presence of meniscus lesions, n (%)		
Yes	136 (49.8)	135 (47.7)
No	137 (50.2)	148 (52.3)
Type of meniscus lesion, n (%)		
No meniscus lesion	137 (50.2)	148 (52.3)
Lateral meniscus	37 (13.6)	43 (15.2)
Medial meniscus	56 (20.5)	56 (19.8)
Both menisci	43 (15.8)	36 (12.7)
Meniscal lesion treatment, n (%)		
Meniscectomy	18 (13.2)	15 (11.1)
Suture	115 (84.6)	113 (83.7)
Left in situ	3 (2.2)	7 (5.2)
Sport type, n (%)		
Pivot/Contact	187 (68.5)	184 (65.0)
Pivot	44 (16.1)	54 (19.1)
No pivot/no contact	6 (2.2)	5 (1.8)
No sport	36 (13.2)	40 (14.1)
Quantitative Pivot Shift-KiRA		
Mean (SD)	3.2 (2.2)	3.3 (2.6)
Side-to-side difference (mm)		
Mean (SD)	6.2 (1.4)	6.4 (1.7)

Data are n (%) or mean (SD). Baseline characteristics were similar between groups, including age subgroup (<25/≥25 years), BMI subgroup (<25 kg/m²/≥25 kg/m²), and type of sport causing the rupture, ensuring balanced randomization. (ACLR, anterior cruciate ligament reconstruction; ALLR, anterolateral ligament reconstruction; SD, standard deviation; BMI, body mass index; KiRA, Knee Rotation Assessment device.) Sex was self-reported by participants. Ethnicity data were not collected in this study.

Table 1: Baseline characteristics of study population.

Secondary outcomes

Tegner activity levels, Lysholm score classes, Subjective KOOS score, and its subscales, were similar between the two groups except for a minor yet statistically significant difference observed in the Quality of Life subscale, which was higher in the ACLR + ALLR group. Detailed results are shown in [Table 2](#) and [Fig. 6](#).

Statistically significant differences, favouring the ACLR + ALLR group, were observed for the subjective IKDC score and the Lysholm score. However, the proportion of patients achieving the MCID (subjective

IKDC: 8.3; Lysholm 11.2) for these two scores did not differ between groups, and post-operative side-to-side laxity was also comparable across both treatment arms ([Table 2](#)).

Overall, no clinically meaningful differences were observed in knee stability ([Table 2](#)), or return-to-sport rates between groups ([Supplementary Appendix](#)).

Post-hoc outcomes

Formal treatment-by-subgroup interaction analyses were performed using logistic regression to evaluate graft failure at 5 years, no interaction was statistically significant ([Supplementary Appendix](#)). When evaluating time to graft failure, only the interaction with Tegner ≥7 versus <7 was statistically significant ($p = 0.0340$), suggesting a better effect of ACLR + ALLR over ACLR in more active patients ([Supplementary Appendix](#)).

The analysis to assess the timing of graft failures in response to concerns that the higher failure rate observed in the BPTB group could be related to early fixation issues due to press-fit femoral tunnel fixation. Early graft failures did not differ between groups: at 1 year (ACLR 1.10% [0.36%–3.37%] versus ACLR + ALLR 1.06% [0.34%–3.25%], $p = 0.96$) and 2 years (4.40% [2.52%–7.61%] versus 1.41% [0.53%–3.72%], $p = 0.07$). A statistically significant difference in failure rates appeared only from 3 years onwards (6.23% [3.92%–9.83%] versus 1.77% [0.74%–4.19%], $p = 0.02$) ([Supplementary Appendix](#)).

Complications reported in this study included both surgical and non-surgical events (S). Non-surgical complications included arthrofibrosis, anterior knee pain, lateral knee pain and patellar tendinopathy. Surgical complications were defined by re-operation for revision ACL reconstruction, cyclops lesions, and meniscal surgery. All other re-operations occurred with a frequency of less than 1%. The total reoperation rate across all participants was 14.8%. Overall, both non-surgical complications and those requiring reoperation were significantly more common in the ACLR group compared to the combined ACLR + ALLR group (21/302, 7.0% versus 67/291, 23.0%, 95% CI: [12.0; 17.7], $p < 0.0001$). In addition to re-operation for graft failure, these differences were primarily driven by a significantly lower incidence of surgeries for cyclops syndrome (3/302, 1.0% versus 24/291, 8.2%, 95% CI: [2.9; 6.2], $p < 0.0001$) and secondary meniscectomy (6/302, 2.0% versus 16/291, 5.5%, 95% CI: [2.2; 5.2], $p = 0.02$) in the ACLR + ALLR group ([Supplementary Appendix](#)). Multivariate logistic regression confirmed that ACLR was associated with a significantly higher odds ratio of reoperation in comparison to ACLR + ALLR (OR 3.91; 95% CI: [2.35–6.76], $p < 0.0001$).

The odds of reoperation was also significantly higher in patients under 25 years old compared to older patients (OR 1.68; 95% CI: [1.04–2.77], $p = 0.035$). Complications not requiring surgery were also significantly

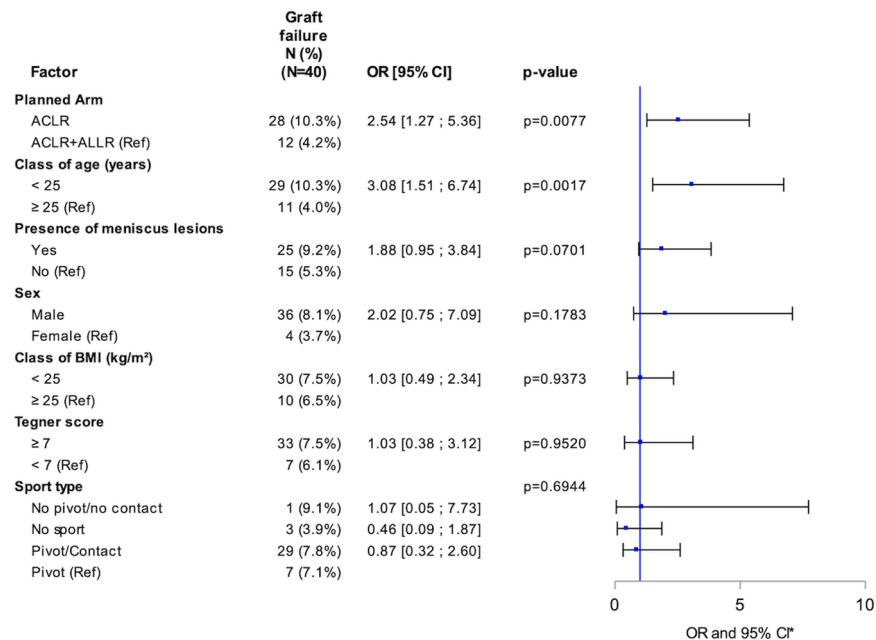


Fig. 3: Forest plot shows odds ratios (ORs) with 95% CIs from a multivariable logistic regression in the Full Analysis Set (n = 556; events = 40). Covariates included planned arm (isolated ACL reconstruction [ACLR] versus ACLR combined with anterolateral ligament reconstruction [ACLR + ALLR, reference]), age (<25 versus ≥25 years [reference]), presence of meniscal lesions (yes versus no [reference]), sex (male versus female [reference]), BMI class (<25 versus ≥25 kg/m² [reference]), preoperative Tegner score (≥7 versus <7 [reference]), and sport type at injury (no pivot/no contact; no sport; pivot/contact versus pivot [reference]). Squares denote point estimates; horizontal bars denote 95% CIs; the vertical line indicates no effect (OR = 1).

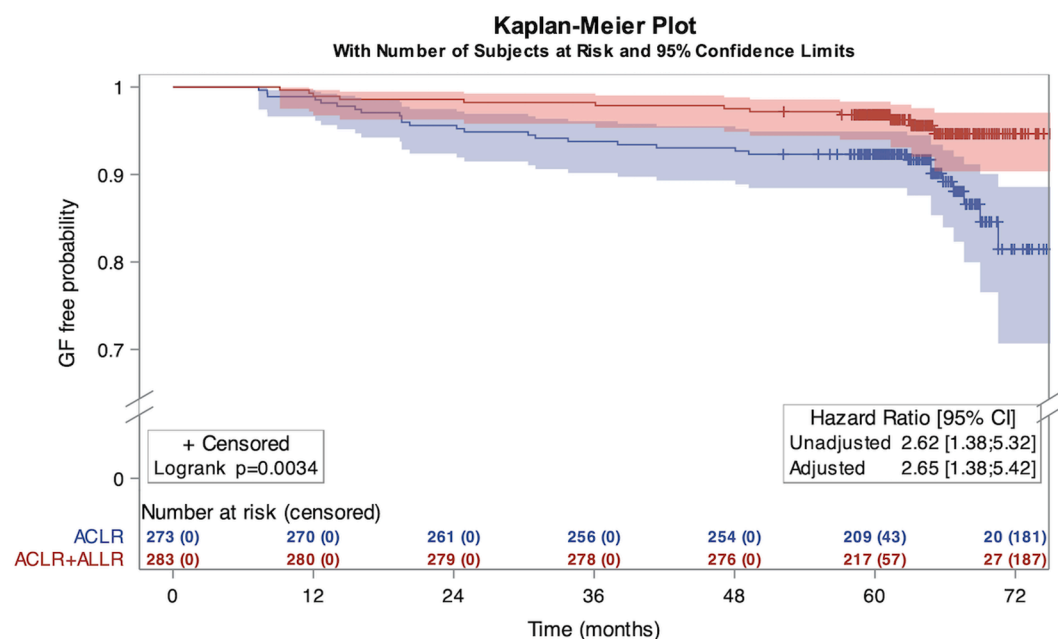


Fig. 4: Kaplan Meier plot. ACLR, anterior cruciate ligament reconstruction; ALLR, anterolateral ligament reconstruction; GF, graft failure. The Kaplan-Meier curves account for censoring after 48 months for patients lost to follow-up or event-free at their last assessment. The overall graft failure rate of 7.2% (40/556) in the FAS reflects observed events at 5 years, consistent with the survivorship estimate.

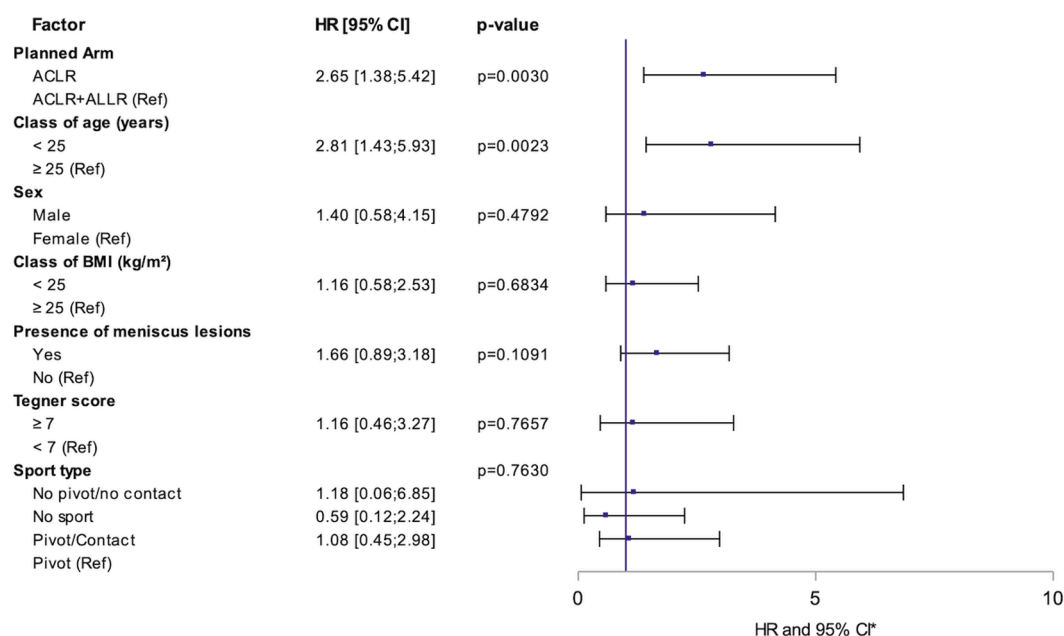


Fig. 5: Adjusted hazard ratios for graft failure over 5 years. Forest plot shows hazard ratios (HRs) with 95% CIs from a Cox proportional hazards model in the Full Analysis Set (n = 556; events = 40). Covariates were planned arm (isolated anterior cruciate ligament reconstruction [ACLR] versus ACLR combined with anterolateral ligament reconstruction [ACLR + ALLR, reference]), age (<25 versus ≥25 years [reference]), sex (male versus female [reference]), body-mass index class (<25 versus ≥25 kg/m² [reference]), presence of meniscal lesions (yes versus no [reference]), preoperative Tegner score (≥7 versus <7 [reference]), and sport type at injury (no pivot/no contact; no sport; pivot/contact versus pivot [reference]). Squares denote point estimates; horizontal bars denote 95% CIs; the vertical line indicates no effect (HR = 1). HR, hazard ratio; CI, confidence interval.

more common in the ACLR group ([Supplementary Appendix](#)).

There was no significant difference in the rate of contralateral ACL rupture between the two groups (ACLR, 8.6% and ACLR + ALLR 9.3%; 95% CI: [6.6; 11.2], p = 0.77) ([Supplementary Appendix](#)).

At 6 months, a greater proportion of patients in the ACLR group achieved a mixed isokinetic strength ratio ≥0.90 compared with ACLR + ALLR (93.4% versus 67.8%; p < 0.0001). By 5 years, return-to-sport outcomes were comparable between groups: overall RTS was 95.3% (97.1% ACLR versus 93.6% ACLR + ALLR; p = 0.066), median time to RTS was 7.9 months in both groups (p = 0.54), and Tegner activity levels at last follow-up were superposable (p = 0.37).

Discussion

The main findings of this randomised controlled trial were that combined reconstructions using hamstring tendon autograft outperformed ACLR with a BPTB autograft with respect to significantly lower graft rupture rates, and overall rates of complications and re-operations. The NNT to prevent one graft rupture was 17 overall, but 9 in young adults under the age of 25. This finding is of particular importance because

BPTB autograft ACLR is widely considered the reference (especially in North America) for young adults, high-risk patients, but this study has demonstrated that even in this specific category, combined reconstructions are advantageous. Although these are new findings, they are broadly compatible with the existing clinical and biomechanical literature demonstrating that combined reconstructions confer improved clinical outcomes and less failure rate.^{17,18}

A previous RCT (STABILITY) and recent meta-analyses of randomised controlled trials also reported a significant reduction in graft failure rates when comparing ACLR with ACLR + LEAP.^{14,17,18} However, that study utilized hamstring tendon autograft ACLR in the control group. The current study therefore provides new findings demonstrating that reduced graft failure rates observed with hamstring ACLR and ALLR hold true even when compared to the commonly used, patellar tendon autograft. In addition, the current study also differs by evaluating an anatomic rather than a non-anatomic LEAP (ALLR) over a considerably longer follow-up period. Additionally, numerous non-randomised comparative studies have shown similar improvement in graft failure rates when combined procedures have been performed.^{9,10,30,31}

Measure	ACLR (n = 273)	ACLR + ALLR (n = 283)	P-value
Preoperative subjective IKDC score, n	272	283	
Mean (SD)	54.5 (14.5)	53.7 (13.7)	
Postoperative subjective IKDC score (5 years), n	243	268	0.04
Mean (SD)	89.4 (10.9)	91.0 (10.3)	
[95% CI]	[88.0; 90.7]	[89.8; 92.3]	
Patients achieving MCID IKDC score, n (%)	225 (92.6)	258 (96.3)	0.07
Preoperative Lysholm score, n	272	281	
Mean (SD)	69.2 (19.0)	68.0 (18.9)	
Postoperative Lysholm score (5 years), n	243	268	0.026
Mean (SD)	89.0 (12.4)	91.1 (11.9)	
[95% CI]	[87.4; 90.6]	[89.6; 92.5]	
Patients achieving MCID Lysholm score, n (%)	160 (65.8)	183 (68.3)	0.56
Class of Lysholm (at 5 years), n(%)	243	268	0.4
Fair (0–64)	14 (5.8)	12 (4.5)	
Good (65–83)	38 (15.6)	33 (12.3)	
Excellent (84–100)	191 (78.6)	223 (83.2)	
Preoperative Tegner score, n	273	283	
Mean (SD)	7.6 (1.5)	7.5 (1.5)	
Preoperative Tegner subgroup, n(%)			
<7	53 (19.4)	62 (21.9)	
≥7	220 (80.6)	221 (78.1)	
Postoperative Tegner (5 years), n	247	269	0.35
Mean (SD)	6.6 (1.6)	6.7 (1.7)	
Preoperative side to side laxity, n	267	273	
Mean (SD)	6.2 (1.5)	6.4 (1.7)	
Postoperative (12 months) side to side laxity, n	139	151	0.17
Mean (SD)	1.2 (1.4)	0.9 (1.0)	

Data are n (%) or mean (SD). P-value on postoperative data: Wilcoxon test for IKDC and Lysholm Score, Student test for Tegner score and side to side laxity and Chi2 test form Class of Lysholm score. P-values are not provided for rows where statistical comparisons were not applicable. This includes categories with zero events, descriptive subgroups, or summary rows aggregating multiple categories. These rows are presented for descriptive purposes only and were not subjected to formal hypothesis testing. (ACLR, anterior cruciate ligament reconstruction; ALLR, anterolateral ligament reconstruction; SD, standard deviation; IKDC, International Knee Documentation Committee; MCID, minimal clinically important difference.)

Table 2: Comparison of pre and postoperative outcomes for all patients.

These clinical results are strongly supported by biomechanical data. Several studies have demonstrated that adding an extra-articular procedure at the time of ACLR better restores knee kinematics.³² Even in the absence of an anterolateral injury, adding a lateral extra-articular procedure has been shown to confer a protective effect against graft failure by load sharing with the ACL graft.³³

The addition of ALLR did not increase the risk of surgical or non-surgical complications in this study or in previous reports.^{12,34,35} In contrast, it was demonstrated that combined ACLR + ALLR was associated with a significant reduction in overall re-operation rates (including for secondary meniscal procedures and stiffness related complications). These findings further reflect that combined procedures better restore normal knee kinematics and confer a protective effect on the repaired meniscus. These findings are also consistent with the previous literature.³⁶ Pioger et al. reported patients who underwent ACLR were more than two-fold more likely to undergo a secondary medial meniscectomy compared with those who underwent combined ACLR + LEAP (hazard ratio, 2.5; 95% CI, 1.5–4.1; $p < 0.001$).³⁷ Similarly, the increased rate of cyclops syndrome in the ACLR group was also expected and has been reported in other series to be predominantly attributed to the use of BPTB grafts which are well known to be associated with increased rates of extension deficit and cyclops syndrome compared to hamstring tendon grafts.³⁸ However, our study's primary objective was to compare two established surgical techniques—ACLR with BPTB autograft versus combined ACLR + ALLR with hamstring autograft—in a high-risk young adults (18–35 years). This pragmatic design reflects real-world practice, not isolating graft or ALLR effects, but demonstrating reduced graft failure

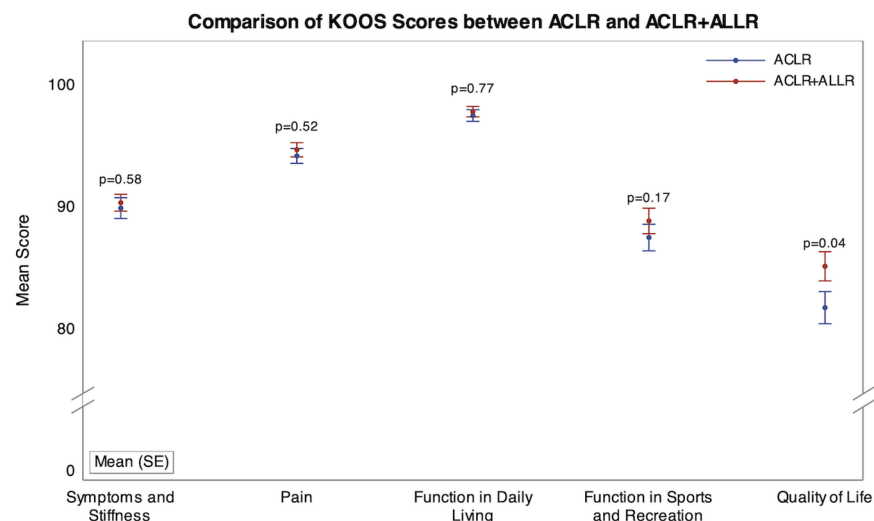


Fig. 6: KOOS Subscales values at 5 years follow-up.

(28/273, 10.3% versus 12/283, 4.2%, $p = 0.006$) and reoperation rates with ACLR + ALLR.

LEAPs were widely used in the 1970s–1980s, initially performed in isolation to address anterolateral rotatory instability without intra-articular ACL reconstruction, leading to overconstraint and poor outcomes, prompting their abandonment. Later, in the 1990s, LEAPs were combined with intra-articular ACL reconstruction to enhance stability, but non-anatomic techniques still risked overconstraint. In contrast, ALLR is a contemporary technique, with the first case series published only in 2015.³¹ It differs significantly from previously abandoned procedures because it is an anatomic procedure that seeks to restore normal knee kinematics, it is percutaneous and does not require harvest of the ITB (which plays a role in anterolateral knee stability). The findings of this RCT demonstrate that ALLR is a safe and effective procedure.

Although ACLR + ALLR outperformed ACLR with respect to graft failure, complication, and overall reoperation rates, there were no clinically important differences regarding other clinical outcomes including knee stability parameters, return to sport rates, and post-operative Tegner activity levels or other PROMS. Although statistically significant differences were observed in some PROMs, these did not exceed minimal clinically important differences (Table 2) and are unlikely to be clinically relevant. Regardless, clinical outcomes in both groups were excellent with high rates of return to sport, and high proportions achieving excellent Lysholm scores.

In this study, contralateral ACL rupture rates were similar between, providing a benchmark for external validity in active young adults (mean 25 years). The approximately 9% rate aligns with prior studies, such as Mohtadi et al. (10% for BPTB at 5 years),³⁹ reflecting our cohort's high sport activity (526/556, 95% frequent sports, 469/556, 84.4% pivot sports). The comparable contralateral rates confirm balanced risk profiles across groups, though the NNT for graft failure prevention is likely higher in low-risk, sedentary populations.

There was a significant interaction between ACLR + ALLR and baseline Tegner activity. These subgroup findings are exploratory and hypothesis-generating, as the trial was powered only for the primary overall effect, not interaction testing, and multiple subgroups were evaluated without multiplicity adjustment. The results suggest that patients with higher baseline activity (Tegner ≥ 7) may derive greater benefit from ACLR + ALLR, with larger relative reductions in graft failure risk. Confirmation of these differential effects requires prospective, adequately powered trials designed specifically to assess treatment effect heterogeneity across these subgroups.

Overall, these findings should be interpreted with consideration to the economic burden of ACL reconstruction. It is reported that the average cost of

ACLR is \$24,707 per procedure with a total annual expenditure of \$3.7 billion in the United States.⁴⁰ With a significant reduction in graft failure, up to four times less than ACLR, ACLR + ALLR offers a significant opportunity for cost savings (especially because there is no additional implant cost to the technique reported in this study) in addition to superior clinical results.

The main limitation of this study is that different ACL graft choices were studied in each group. However, the choice to use different grafts was intentional and firstly based on the rationale of comparing against the commonly used technique, secondly based on existing evidence suggesting that combined the results of ACLR & ALLR performed with hamstring tendons would be superior, and thirdly because using hamstring autografts for the combined procedure does not require any additional graft harvest. Another limitation of this study is the use of different fixation methods (press-fit for ACLR versus interference screws for ACLR + ALLR). Although systematic reviews indicate comparable failure rates between these techniques,⁴¹ early graft failures, typically linked to fixation, showed no differences within the first 2 years in our RCT. The divergence in failure rates beyond 3 years suggests that outcomes are more likely influenced by the protective effect of lateral extra-articular procedures rather than fixation methods. Contemporary biomechanical data also support equivalence between press-fit and interference screw fixation at time zero.⁴² Further limitations of this study are the single centre design. Three very experienced surgeons (respectively 600, 400 and 300 ACL per year for more than 12 years) were involved which may aid generalizability but it is also recognized that as a specialist centre more complex and higher risk cases are likely to be encountered. However, randomisation was not stratified by surgeon and the analyses did not adjust for clustering by operator; therefore, an unmeasured surgeon effect cannot be excluded. An objective measure of a populations risk for re-injury is based on the contralateral ACL injury rate. The original protocol and [ClinicalTrials.gov](https://www.clinicaltrials.gov/ct2/show/study?term=NCT03740022) registry (NCT03740022) specified a 3-year primary endpoint for graft failure, but the follow-up was extended to 5 years due to the impact of the COVID-19 pandemic, which disrupted surgical practices, patient inclusion, and return-to-sport timelines. The 3-year data were available but not published separately, as preliminary 1-year findings were reported,³⁵ and to avoid redundancy because mid-term 5-year outcomes provide stronger clinical relevance. The mean time from injury to surgery (around 7 months), influenced by France's healthcare system, classifies this study as evaluating chronic ACL tears (>3–6 months post-injury). This may limit extrapolation to acute tears (<6 weeks), where outcomes, such as secondary meniscal lesions, may differ due to shorter injury duration.

In conclusion, combined ACLR + ALLR using hamstrings tendon autografts reduced graft failure rates between 3 and 4 fold when compared to ACLR using a BPTB autograft. The NNT were 17 overall, but 9 in young adults (age <25) for whom BPTB grafts are most strongly advocated due to the elevated risk of graft rupture in this population. Patients undergoing ACLR + ALLR had had significantly fewer overall reoperations and complications while postoperative laxity PROMs were equivalent. These findings support combined reconstructions in high-risk patients.

Contributors

Alessandro Carrozzo, Thais Dutra Vieira, and Damien de Paulis accessed and verified the underlying data. All authors had full access to all the data in the study. All authors were involved in and took responsibility for the decision to submit the manuscript for publication.

Bertrand Sonnerly-Cottet, Jean-Marie Fayard, Benjamin Freychet, Mathieu Thauinat: Investigation (surgeons who included patients), Conceptualisation, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualisation, Writing—original draft, Writing—review & editing.

Thais Dutra Vieira: Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualisation, Writing—original draft, Writing—review & editing.

Alessandro Carrozzo, Hervé Poilvache: Resources, Software, Supervision, Validation, Visualisation, Writing—original draft, Writing—review & editing.

Adnan Saithna: Supervision, Validation, Visualisation, Writing—original draft, Writing—review & editing.

The collaborator group was involved in patient follow-up, clinical evaluation, and the collection, registration, and management of study data.

Data sharing statement

Individual participant data that underlie the results reported in this article, after de-identification (text, tables, figures, and appendices), will be made available, along with the study protocol. Data will be accessible from 9 months to 36 months following publication to investigators whose proposed use has been approved by an independent review committee (“learned intermediary”) for the purpose of individual participant data meta-analyses. Proposals for data access may be submitted up to 36 months after publication. After this period, the data will remain available in our university’s data warehouse, though without investigator support beyond the deposited metadata. Information on submitting proposals and accessing the data can be obtained by contacting arc@orthosanty.fr.

Declaration of interests

Alessandro Carrozzo reports a fellowship grant from Arthrex. Adnan Saithna, Bertrand Sonnerly-Cottet, Jean-Marie Fayard, and Mathieu Thauinat report consulting fees from Arthrex, with Sonnerly-Cottet, Fayard, and Thauinat also receiving research grants from Arthrex. Sonnerly-Cottet reports equity ownership in AREAS, and Fayard reports consulting fees from Newclip Techniques. Benjamin Freychet, Hervé Poilvache, and Thais Dutra Vieira have nothing to disclose.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.j.lanepe.2025.101561>.

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